

SWP 4: Nested Logit Demand Estimation for the Cola Market - Meng Yang 640815

1. Introduction

This study extends SWP 3 by addressing a critical limitation: the Independence of Irrelevant Alternatives (IIA) restriction. SWP 3 estimated two separate non-nested logit models—a brand-level model (elasticity -1.40) and a 12-can package model (-2.71)—that forced symmetric substitution patterns. This contradicts intuitive dietary segmentation, as consumers likely substitute more readily within dietary categories (regular vs. diet) than across them.

The fundamental problem is that IIA restricts cross-price elasticities to be proportional to market shares regardless of product similarity. In reality, consumers likely substitute more readily within dietary categories (regular vs. diet) than across them, and within package formats (bottles vs. cans) depending on consumption occasion. SWP 3 could not capture these nuanced substitution patterns because the aggregate logit specification treats all products as equally differentiated.

This paper introduces an aggregate nested logit demand model combining brand and package dimensions into 8 inside goods (4 brands $\times$ 2 formats: 2L bottles and 12-can packs). The nesting structure relaxes IIA by dietary status (Regular vs. Diet nests), enabling asymmetric substitution patterns. The central research question examines how this nested structure affects price elasticities and whether cross-price elasticities exhibit expected within-nest versus across-nest asymmetry.

2. Data and Market Definition

2.1 Data Source and Sample

The analysis draws on IRI scanner data retrieved from the HU-Box repository, covering the period 2004–2006. The primary dataset (cola.coke.pepsi.csv) contains product-store-week level observations with information on prices, volumes, promotional activities, and trip counts. Supplementary files provide store characteristics, cost shifters (aluminum prices, sugar prices, gasoline prices), and instrumental variables.

2.2 Products: 8 Inside Goods

The full IRI dataset contains 75,317 store-week observations across all cola SKUs (SWP 3). To focus on the strategically most important segments and maintain computational tractability for the nested logit estimation, this analysis restricts attention to the two dominant package formats: 2-liter bottles (immediate consumption, family occasions) and 12-can multipacks (stock-up purchases) which can create a product space that captures both brand loyalty and format-specific consumption occasions (immediate consumption for bottles, stockpiling for cans).

Caffeine-free variants are excluded as they represent less than 3% of category sales and exhibit systematically different demand patterns see in SWP 3.

The final estimation sample contains 15,685 observations with 25 variables after merging product attributes, market shares, instruments, and nest definitions.

2.3 Market Shares

The outside good dominates with 93% mean share, motivating the discrete choice framework. Even at peak cola consumption, 30% of trips exclude cola, requiring explicit outside good modeling.

3. Theoretical Model: Nested Logit Demand

3.1 Utility Specification

The nested logit extends the aggregate logit from SWP 3 by allowing correlation in unobserved preferences within predefined nests. For consumer i choosing product j in nest g at market t , indirect utility is:

$$u_{ijt} = \delta_{jt} + \zeta_{ig}(\sigma) + (1 - \sigma)\epsilon_{ijt}$$

$$\delta_{jt} = \alpha p_{jt} + \sum_b \beta_b D_j^b + \gamma' X_{jt} + \xi_{jt}$$

Where function δ_{jt} represents mean utility (price, brand×format fixed effects, promotions, store fixed effects), $\zeta_{ig}(\sigma)$ is a nest-specific random component with nesting parameter $\sigma \in [0,1]$, and ϵ_{ijt} is IID Type-I Extreme Value. The correlation within nests is $\rho = 1 - \sigma^2$

3.2 Choice Probabilities and Share Equations

The nested logit yields a two-level choice structure. Within-nest choice probability (conditional on choosing nest g):

$$s_{j|g,t} = \frac{\exp(\delta_{jt}/(1-\sigma))}{\sum_{k \in g} \exp(\delta_{kt}/(1-\sigma))}$$

Across-nest choice probability (nest selection):

$$s_{g,t} = \frac{\exp((1-\sigma)I_{gt})}{1 + \sum_h \exp((1-\sigma)I_{ht})}$$

$$I_{gt} = \ln \sum_{k \in g} \exp(\delta_{kt}/(1-\sigma))$$

Where I_{gt} is the inclusive value capturing nest attractiveness. Overall market share represent as:

$$s_{jt} = s_{j|g,t} \times s_{g,t}$$

The outside good (non-cola alternatives) has normalized utility $\delta_0 = 0$, yielding

$$s_{0t} = 1 / (1 + \sum_h \exp((1 - \sigma)I_{ht}))$$

The nested logit model categorizes products into REGULAR and DIET nests-each containing Coke and Pepsi in both bottle and can formats-with each product holding an equal average within-nest share of 25% and each nest capturing an average of 3.5% of the overall market.

Table 1: Market Share

Statistic	Sj // g	ln(Sj // g)	Sg (Nest Share)
Min	0.0013	-6.66	0.0027
1st Quartile	0.0725	-2.62	0.0164
Median	0.1746	-1.75	0.0280
Mean	0.2504	-1.86	0.0346
3rd Quartile	0.3855	-0.95	0.0439
Max	0.9771	-0.02	0.5164

The substantial variation in within-group shares ($CV \approx 1.0$) and the range of $\ln(Sj // g)$ from -6.7 to -0.02 indicate meaningful heterogeneity in conditional choice probabilities, justifying the nested structure.

Since SWP 3 revealed IIA violations: cross-elasticities between diet and regular variants were implausibly high, while within-diet substitution was understated. Health-conscious consumers exhibit stronger substitution within their dietary category than across. The σ parameter captures this correlation-if $\sigma \rightarrow 0$, the model collapses to standard logit (no correlation); if $\sigma \rightarrow 1$, products within nests become perfect substitutes.

3.4 Estimating Equation

Applying the Berry (1994) inversion, the linear estimating equation is:

$$\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt} + \sigma \ln(s_{j|g,t})$$

Substituting δ_{jt} :

$$\underbrace{\ln(s_{jt}) - \ln(s_{0t})}_{\text{Log-odds vs. outside good}} = \alpha p_{jt} + \sum_b \beta_b D_j^b + \gamma' X_{jt} + \underbrace{\sigma \ln(s_{j|g,t})}_{\text{Endogenous!}} + \xi_{jt}$$

Critical distinction from SWP 3: The term $\ln(s_{j|g,t})$ appears on the right-hand side but is endogenous-correlated with ξ_{jt} through simultaneous determination of within-nest

shares. This necessitates instrumental variables (BLP-style instruments: characteristics of other products in the same nest) for consistent estimation of σ .

The nesting parameter σ is identified from exogenous variation in nest composition: when rivals in the same diet category change prices or promotions, the within-group share $\ln(s_{j|g,t})$ shifts for reasons uncorrelated with product j 's unobserved quality ξ_{jt} .

4. Econometric Strategy & Endogeneity Correction

4.1 Endogeneity Problems

The nested logit specification introduces two endogenous variables requiring instrumental variable correction:

The model faces two key endogeneity concerns. First, prices p_{jt} are correlated with unobserved demand shocks, biasing the price coefficient (α) toward zero in OLS. Second, the within-group share $\ln(s_{j|g,t})$ is correlated with unobserved nest preferences, biasing the nesting parameter (σ) upward and overstating within-nest loyalty. Failure to instrument $\ln(s_{j|g,t})$ produces inconsistent estimates of the nesting parameter, overstating within-nest correlation and distorting substitution patterns.

4.2 Instrumental Variable Strategy

Three instrument classes address distinct sources of exogenous variation:

To address endogeneity, the model employs three instrumental variable sets-Hausman-type (prices in other markets), cost shifters (commodity prices and their interactions), and BLP-style (rival product characteristics within the same nest)-which satisfy the exclusion restriction by influencing local demand only through supply-side channels or competitive structure, while remaining strongly relevant due to cost pass-through, differential input exposure, and within-nest competitive dynamics.

First-stage equations:

$$p_{jt} = \pi_1^p \text{Hausman}_t + \pi_2^p \text{Cost}_t + \pi_3^p \text{BLP}_t + \gamma' W_{jt} + \nu_{jt}^p$$

$$\ln(s_{j|g,t}) = \pi_1^s \text{BLP}_t + \pi_2^s \text{Hausman}_t + \gamma' W_{jt} + \nu_{jt}^s$$

where W_{jt} includes all exogenous regressors (brand×format fixed effects, store dummies, own promotions).

4.3 Identification and Diagnostics

The exclusion restrictions rely on:

- Hausman instruments: Geographically separate markets face common wholesale costs but independent demand shocks
- Cost shifters: Global aluminum/sugar markets are sufficiently large that local cola demand has negligible price impact; interactions (aluminum_x_can, sugar_x_regular) provide differential instruments for package-specific and diet-specific cost variation
- BLP instruments: Following Berry, Levinsohn, and Pakes (1995), the sum of rival characteristics in the same nest affects j 's conditional share through competitive substitution but is uncorrelated with j 's unobserved quality ξ_{jt} .

Diagnostic tests:

- Weak instruments: First-stage F-statistics > 10 (Stock-Yogo critical value)
- Endogeneity: Wu-Hausman test rejects OLS consistency if χ^2 significant
- Overidentification: Sargan-Hansen J-test validates instrument exclusion restrictions when overidentified

The IV strategy thus combines spatial variation (Hausman), cost-driven variation (commodity prices), and market structure variation (BLP) to identify both price sensitivity α and the nesting parameter σ .

5. Estimation Results: OLS Benchmarks

5.1 OLS Aggregate Logit (No Nesting)

The baseline OLS aggregate logit ignores nesting, treating all 8 products as equally differentiated. Results show in Table 2 that Price coefficient = -4.347 (SE: 0.072, $t = -60.6$), $R^2 = 0.9805$. This implies implausibly high elasticity (~ -4.8), suggesting omitted variable bias from ignoring nest structure. Brand $\times$ format fixed effects show expected hierarchy: bottle formats more negative than cans (e.g., Coke Classic Bottle: -3.68 vs. Can: -2.08). See Appendix A.

5.2 OLS Nested Logit (biased - no IV)

Adding the endogenous within-group share, results shown in Table 2: $\sigma = 0.686$ (SE: 0.006, $t = 112.1$), price = -1.653 (SE: 0.059), $R^2 = 0.9892$. The σ indicates substantial within-nest correlation ($\rho = 1 - 0.686^2 = 0.53$), confirming nesting relevance. However, σ is biased upward— $\ln(s_j|g)$ correlates with unobserved demand shocks. Price coefficient attenuates toward plausible range but remains inconsistent. Promotional effects shrink (display: 0.46 $\rightarrow$ 0.11), suggesting OLS aggregate conflated nest shocks with promotion effectiveness. See Appendix B: OLS Nested Logit (biased - no IV)

Table 2: OLS Results

$\sigma = \ln(s_{-j g})$	(1) Aggregate	(2) Nested
(within-group share)	-	0.686 (0.006)

Price	-4.347 (0.072)	-1.653 (0.059)
Display	0.463 (0.016)	0.113 (0.012)
Feature	0.231 (0.024)	0.202 (0.018)
Coupon	1.152 (0.053)	0.666 (0.039)
R ²	0.9805	0.9892
N	15685	15685

Standard errors in parentheses. Both include brand×format FE and 12 store dummies. Column (2) σ is biased due to endogeneity. The R² increase in (2) reflects spurious precision-the endogenous regressor captures variation it should not. IV correction addresses both biases simultaneously.

6. IV Results: Main Specification

6.1 IV Estimation: Aggregate vs. Nested Logit

Two IV specifications are estimated: (1) IV Aggregate Logit instrumenting price only, ignoring nesting; and (2) IV Nested Logit instrumenting both price and $\ln(s_{j|g})$, which is preferred.

Table 3: IV Estimation Results

	(1) IV Aggregate Logit (Price Only)	(2) IV Nested (Both IVed) - Preferred
$\sigma = \ln(s_{j g})$	(imposed 0)	0.124 (0.037)***
Price per liter	-11.763 (0.459)***	-9.642 (0.670)***
Display	-0.117(0.0410)*	-0.051 (0.035)***
Feature	-1.301 (0.0997)***	-0.968 (0.115)**
Coupon	0.685 (0.108)***	0.700 (0.092)***
R-squared	0.967	0.9752
Observations	15685	15685

The preferred IV nested model (Column 2) corrects the severe biases in the aggregate specification-where omitted nesting inflated price sensitivity and promotion effects-by instrumenting for both price and within-group share using Hausman LOOC and BLP instruments. The extremely low σ (0.124) implying $\rho = 0.985$ suggests near-perfect substitution within diet categories. While this validates the nesting structure qualitatively, the magnitude may reflect IV overcorrection or weak instrument bias in the presence of many instruments. The OLS estimate ($\sigma = 0.686$, $\rho = 0.53$) may represent a more conservative lower bound on within-nest differentiation.

6.2 Diagnostic Tests

Table 4: IV Diagnostics

Test	(1) IV Aggregate(Nest share)	(2) IV Nested(for Price)
Weak instruments		

- Price equation	F = 239.7***	F = 769.0***
- $\ln(s_{ijl})$ equation	---	F = 168.3***
Wu-Hausman	$\chi^2 = 506.3***$	$\chi^2 = 353.1***$
Sargan over-ID	$\chi^2 = 157.3***$	$\chi^2 = 199.4***$

Note on overidentification: Sargan rejection in both models suggests some instrument correlation with structural errors, possibly from correlation between LOOC price and BLP instruments. However, first-stage F-statistics exceed Stock-Yogo critical values (16.38 for 10% maximal IV size), ensuring reliable inference despite overidentification concerns. The IV nested model is strongly preferred. It produces economically valid parameters, valid $\sigma \in [0,1]$, and superior fit ($R^2 = 0.9752$), while addressing the endogeneity that biases OLS and the misspecification that distorts IV aggregate.

7. Elasticity Computation & Results

This section presents the elasticity estimates derived from the nested logit demand models. We first discuss the parameter estimates that govern substitution patterns, then examine own-price elasticities across products and formats, and finally analyze cross-price elasticities to understand competitive dynamics within and across dietary segments.

7.1 Model Parameters and Summary Statistics

Table 5 reports the key parameter estimates and summary statistics for both the OLS and IV nested logit specifications. The IV model represents our preferred specification as it corrects for the endogeneity of price and the within-group share using LOOC price, cost-shifter, and BLP instruments.

Table 5: Parameter Estimates and Market Summary

Parameter / Statistic	OLS Nested Logit	IV Nested Logit
Structural Parameters		
σ (nesting parameter)	0.686	0.124
α (price coefficient)	-1.653	-9.642
Nest Shares		
Group 1 (REGULAR)	-	0.0076
Group 2 (DIET)	-	0.0056
Outside Good	0.994	0.96
Representative Prices (\$/liter)		
Bottle formats (all brands)	0.613–0.630	0.613–0.630
Can formats (all brands)	0.836–0.873	0.836–0.873

The IV model produces a substantially lower nesting parameter (0.124 vs. 0.686), indicating extremely strong within-group correlation ($\rho = 1 - 0.124^2 = 0.985$) far exceeding OLS estimates. This confirms that consumers view products within the same diet category as near-perfect substitutes. The price coefficient is nearly six times larger in absolute value in the IV specification (−9.642 vs. −1.653), reflecting severe downward bias in OLS when both price endogeneity and nest share endogeneity are ignored. The outside good share increases from 99.4% (OLS) to 96% (IV), indicating that OLS understated the relevance of cola products in consumer choice due to attenuation bias.

7.2 Own-Price and Cross-Price Elasticities

Table 6 presents OLS Nested Logit Elasticities for all eight products under both specifications.

Table 6: OLS Nested Logit Elasticities

Product	Own-Price Elasticity	Cross-Price Elasticity (Within Group)	Cross-Price Elasticity (Across Groups)
Bottle Formats			
Coke Classic Bottle	-2.745	0.484	0.005
Diet Coke Bottle	-2.791	0.445	0.003
Pepsi Bottle	-2.96	0.359	0.004
Diet Pepsi Bottle	-3	0.288	0.002
Can Formats			
Coke Classic Can	-3.197	1.211	0.012
Diet Coke Can	-3.05	1.359	0.01
Pepsi Can	-3.866	0.739	0.007
Diet Pepsi Can	-3.843	0.722	0.005

The OLS nested logit reveals that can formats are more price-elastic than bottles (−3.0 to −3.9 vs. −2.7 to −3.0), confirming stockpiling behavior where consumers actively compare prices for multipacks. Within-nest cross-elasticities are 50–100× larger than across-nest elasticities (e.g., 0.484 vs. 0.005 for Coke Classic Bottle), demonstrating strong substitution within diet categories but minimal cross-diet switching. However, the upward-biased σ (0.686) likely understates true within-nest correlation, as the endogenous $\ln(s_{j|g})$ captures demand shocks rather than true structural correlation, making IV correction essential.

Table 7 : IV Nested Logit Elasticities (Preferred)

Product	Within-Group Share	Own-Price Elasticity	Cross-Price Elasticity (Within Group)	Cross-Price Elasticity (Across Groups)
Bottle Formats				

Coke Classic Bottle	0.22	-6.548	0.194	0.01
Diet Coke Bottle	0.203	-6.579	0.177	0.007
Pepsi Bottle	0.149	-6.793	0.135	0.007
Diet Pepsi Bottle	0.117	-6.756	0.104	0.004
Can Formats				
Coke Classic Can	0.416	-8.701	0.499	0.025
Diet Coke Can	0.465	-8.652	0.551	0.021
Pepsi Can	0.215	-9.341	0.27	0.014
Diet Pepsi Can	0.215	-9.266	0.263	0.01

The IV correction dramatically increases own-price elasticities to -6.5 to -9.3 , revealing severe attenuation bias in OLS. Can formats remain more elastic than bottles, but the gap widens: cans are now 33% more elastic than bottles versus 15% in OLS. Within-nest cross-elasticities are 20–50 $\times$ larger than across-nest elasticities (e.g., 0.194 vs. 0.010 for Coke Classic Bottle), validating the nested structure. However, the extreme price sensitivity (-9.3 for Pepsi Can) and near-perfect within-nest substitution (implied by $\sigma = 0.124$) suggest the model may approach the boundary of economic plausibility. Within-group shares differ dramatically between formats and brands. In the IV specification, Coke Classic and Diet Coke Cans capture 41.6% and 46.5% of their respective nests—nearly half the diet can segment—while their bottle counterparts hold only 22.0% and 20.3%. This concentration explains why IV cross-elasticities for cans approach 0.5–0.55: high $s_{j|g}$ (0.42–0.47) combined with large $|\alpha|$ (9.64) and $\sigma/(1-\sigma) = 0.14$ multiplies substitution effects. Pepsi and Diet Pepsi show more balanced shares across formats (~ 15 –22%), yielding more moderate cross-elasticities (0.26–0.27).

7.3 Key Findings

Cross-price elasticities reveal the nesting structure. The IV nested logit shows within-nest cross-elasticities of 0.10–0.55, while across-nest elasticities remain near-zero at 0.004–0.025. This 10–50 $\times$ asymmetry validates the diet-based nesting structure, confirming that consumers substitute readily within Regular or Diet categories but rarely cross the diet boundary. The IV correction preserves this asymmetric pattern while dramatically increasing the magnitude of all elasticities.

The extremely low σ (0.124) implying $\rho = 0.985$ suggests near-perfect substitution within diet categories. While this validates the nesting structure qualitatively, the magnitude may reflect IV overcorrection or the presence of many instruments. The OLS estimate ($\sigma = 0.686$, $\rho = 0.53$) may represent a more conservative lower bound on within-nest differentiation.

7.4 Comparison with SWP 3

The IV nested logit represents a methodological advance over SWP 3's non-nested specifications, but with nuanced performance. Compared to the SWP 3 brand-level

model (mean elasticity -1.40 , symmetric cross-elasticities 0.005 – 0.007), the IV nested model produces substantially higher own-price elasticities (-6.5 to -9.3) that better reflect the combined brand-format market, and crucially generates the asymmetric substitution pattern that IIA-restricted models cannot capture.

Relative to the SWP 3 12-can model (elasticity -2.71), the IV nested model increases can elasticities upward to -8.7 to -9.3 , revealing severe attenuation bias in both SWP 3 specifications. The near-perfect substitution within nests ($\sigma = 0.124$, $\rho = 0.985$) generates economically extreme substitution intensity that should be interpreted with caution.

The IV nested model achieves the qualitative breakthrough of asymmetric substitution but with quantitative challenges: elasticities exceeding -9.0 may overstate true price sensitivity, suggesting that a single nesting parameter may be insufficient for formats with divergent consumer behaviors, or that further instrument pruning is warranted to reduce overidentification bias.

8 Concluding Summary

The nested logit framework successfully addresses SWP 3's IIA restriction, generating asymmetric substitution patterns that validate dietary market segmentation. The IV-estimated nesting parameter $\sigma = 0.124$ implies near-perfect within-nest correlation ($\rho = 0.985$), with mean own-price elasticities of -7.83 ranging from -6.55 (bottles) to -9.34 (cans). These extreme estimates suggest either genuine high sensitivity or IV overcorrection, warranting cautious interpretation. The central managerial insight remains robust: dietary status dominates substitution, cans are 33% more elastic than bottles, and format-specific pricing strategies are essential for profit maximization. Future research should implement random coefficients to relax the restrictive nesting structure. The central managerial insight remains robust: dietary status dominates consumer substitution, and format-specific pricing strategies are essential for profit maximization. Future research should implement random coefficients to relax the restrictive single-parameter nesting structure and validate these findings in a more flexible demand framework.

Appendix:

Appendix A: OLS Aggregate Logit (No Nesting)

```
lm(formula = y ~ -1 + PACKAGE:L5 + store01 + store02 + store03 +  
  store04 + store05 + store06 + store07 + store08 + store09 +  
  store10 + store11 + store12 + price.per.liter + display +  
  feature + coupon, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.9477	-0.4841	-0.0256	0.4483	3.5724

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
store01	1.19642	0.03512	34.071	< 2e-16 ***
store02	-0.07111	0.03188	-2.230	0.0257 *
store03	0.37793	0.03222	11.728	< 2e-16 ***
store04	1.20306	0.03381	35.578	< 2e-16 ***
store05	1.38609	0.03475	39.885	< 2e-16 ***
store06	0.01477	0.03241	0.456	0.6485
store07	0.96310	0.03594	26.794	< 2e-16 ***
store08	0.21621	0.03267	6.618	3.75e-11 ***
store09	0.78985	0.03586	22.024	< 2e-16 ***
store10	1.25369	0.03500	35.820	< 2e-16 ***
store11	1.37884	0.03375	40.857	< 2e-16 ***
store12	0.44863	0.03255	13.784	< 2e-16 ***
price.per.liter	-4.34710	0.07175	-60.586	< 2e-16 ***
display	0.46338	0.01603	28.911	< 2e-16 ***
feature	0.23086	0.02396	9.637	< 2e-16 ***
coupon	1.15244	0.05257	21.921	< 2e-16 ***
PACKAGEBOTTLE:L5COKE CLASSIC	-3.68063	0.05007	-73.510	< 2e-16 ***
PACKAGECAN:L5COKE CLASSIC	-2.08417	0.06397	-32.579	< 2e-16 ***
PACKAGEBOTTLE:L5DIET COKE	-4.05489	0.05020	-80.775	< 2e-16 ***
PACKAGECAN:L5DIET COKE	-2.26003	0.06393	-35.352	< 2e-16 ***
PACKAGEBOTTLE:L5DIET PEPSI	-4.47415	0.04981	-89.828	< 2e-16 ***
PACKAGECAN:L5DIET PEPSI	-2.80138	0.06472	-43.287	< 2e-16 ***
PACKAGEBOTTLE:L5PEPSI	-3.94679	0.05055	-78.081	< 2e-16 ***
PACKAGECAN:L5PEPSI	-2.45352	0.06502	-37.734	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7753 on 15661 degrees of freedom

Multiple R-squared: 0.9805, Adjusted R-squared: 0.9804

F-statistic: 3.277e+04 on 24 and 15661 DF, p-value: < 2.2e-16

Appendix B: OLS Nested Logit (biased - no IV)

```
[1] "=== OLS Nested Logit (biased - no IV) ==="
```

```
> summary(ols_nl)
```

Call:

```
lm(formula = y ~ ln_share_g - 1 + PACKAGE:L5 + store01 + store02 +  
  store03 + store04 + store05 + store06 + store07 + store08 +  
  store09 + store10 + store11 + store12 + price.per.liter +  
  display + feature + coupon, data = dat)
```

Residuals:

```
   Min      1Q  Median      3Q      Max  
-2.0843 -0.3582 -0.0068  0.3459  3.4820
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
ln_share_g	0.686442	0.006126	112.051	< 2e-16 ***
store01	0.625241	0.026654	23.458	< 2e-16 ***
store02	-0.568693	0.024166	-23.533	< 2e-16 ***
store03	0.117462	0.024120	4.870	1.13e-06 ***
store04	0.478708	0.026009	18.406	< 2e-16 ***
store05	0.589734	0.026848	21.965	< 2e-16 ***
store06	-0.384617	0.024409	-15.757	< 2e-16 ***
store07	-0.033214	0.028216	-1.177	0.239
store08	-0.215699	0.024641	-8.754	< 2e-16 ***
store09	-0.228996	0.028223	-8.114	5.27e-16 ***
store10	0.585340	0.026749	21.883	< 2e-16 ***
store11	0.646486	0.025978	24.886	< 2e-16 ***
store12	0.231736	0.024325	9.527	< 2e-16 ***
price.per.liter	-1.653261	0.058613	-28.206	< 2e-16 ***
display	0.112628	0.012345	9.124	< 2e-16 ***
feature	0.202161	0.017849	11.326	< 2e-16 ***
coupon	0.666431	0.039407	16.911	< 2e-16 ***
PACKAGEBOTTLE:L5COKE CLASSIC	-3.234693	0.037514	-86.226	< 2e-16 ***
PACKAGECAN:L5COKE CLASSIC	-2.675613	0.047953	-55.797	< 2e-16 ***
PACKAGEBOTTLE:L5DIET COKE	-3.557256	0.037662	-94.451	< 2e-16 ***
PACKAGECAN:L5DIET COKE	-2.936501	0.048010	-61.165	< 2e-16 ***
PACKAGEBOTTLE:L5DIET PEPSI	-3.683452	0.037773	-97.516	< 2e-16 ***
PACKAGECAN:L5DIET PEPSI	-3.096655	0.048286	-64.131	< 2e-16 ***
PACKAGEBOTTLE:L5PEPSI	-3.308865	0.038087	-86.877	< 2e-16 ***
PACKAGECAN:L5PEPSI	-2.782429	0.048531	-57.333	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5776 on 15660 degrees of freedom

Multiple R-squared: 0.9892, Adjusted R-squared: 0.9891

F-statistic: 5.717e+04 on 25 and 15660 DF, p-value: < 2.2e-16

```
ols_agg$coefficients["price.per.liter"] # Price coef (biased) price.per.liter -4.347099  
> summary(ols_agg)$r.squared 0.9804738
```

Appendix C: IV Aggregate Logit (price instrumented)

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
store01	2.389948	0.084822	28.1759	< 2.2e-16 ***
store02	0.446009	0.048843	9.1314	< 2.2e-16 ***
store03	0.996985	0.056468	17.6558	< 2.2e-16 ***
store04	2.477244	0.087452	28.3269	< 2.2e-16 ***
store05	2.853251	0.098122	29.0785	< 2.2e-16 ***
store06	0.398650	0.048218	8.2676	< 2.2e-16 ***
store07	2.647259	0.111608	23.7192	< 2.2e-16 ***
store08	1.123346	0.072038	15.5937	< 2.2e-16 ***
store09	2.490084	0.111346	22.3634	< 2.2e-16 ***
store10	2.426978	0.084794	28.6219	< 2.2e-16 ***
store11	2.621836	0.085393	30.7033	< 2.2e-16 ***
store12	1.141084	0.061261	18.6266	< 2.2e-16 ***
price.per.liter	-11.762798	0.459124	-25.6201	< 2.2e-16 ***
display	-0.115740	0.040956	-2.8259	0.00472 **
feature	-1.300957	0.099712	-13.0471	< 2.2e-16 ***
coupon	0.684998	0.107543	6.3695	1.949e-10 ***
PACKAGEBOTTLE:L5COKE CLASSIC	0.566645	0.266038	2.1299	0.03319 *
PACKAGECAN:L5COKE CLASSIC	3.832507	0.368995	10.3863	< 2.2e-16 ***
PACKAGEBOTTLE:L5DIET COKE	0.205049	0.266696	0.7688	0.44199
PACKAGECAN:L5DIET COKE	3.651786	0.368852	9.9004	< 2.2e-16 ***
PACKAGEBOTTLE:L5DIET PEPSI	-0.243474	0.264224	-0.9215	0.35682
PACKAGECAN:L5DIET PEPSI	3.214065	0.374386	8.5849	< 2.2e-16 ***
PACKAGEBOTTLE:L5PEPSI	0.367352	0.269805	1.3615	0.17336
PACKAGECAN:L5PEPSI	3.600315	0.377584	9.5351	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(iv_agg, diagnostics = TRUE)

```
> print(summary(iv_agg, diagnostics = TRUE))
```

Call:

```
ivreg(formula = y ~ -1 + PACKAGE:L5 + store01 + store02 + store03 +
  store04 + store05 + store06 + store07 + store08 + store09 +
  store10 + store11 + store12 + price.per.liter + display +
  feature + coupon | -1 + PACKAGE:L5 + store01 + store02 +
  store03 + store04 + store05 + store06 + store07 + store08 +
  store09 + store10 + store11 + store12 + display + feature +
  coupon + avg_price_other_stores_brand + aluminium.price +
  sugar.price, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.768507	-0.613316	0.006026	0.595014	4.319056

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
store01	2.38995	0.08342	28.648	< 2e-16 ***
store02	0.44601	0.05126	8.702	< 2e-16 ***
store03	0.99699	0.05533	18.020	< 2e-16 ***
store04	2.47724	0.08655	28.622	< 2e-16 ***
store05	2.85325	0.09702	29.408	< 2e-16 ***
store06	0.39865	0.04767	8.363	< 2e-16 ***
store07	2.64726	0.10909	24.267	< 2e-16 ***
store08	1.12335	0.06795	16.532	< 2e-16 ***
store09	2.49008	0.10990	22.658	< 2e-16 ***
store10	2.42698	0.08235	29.472	< 2e-16 ***
store11	2.62184	0.08494	30.868	< 2e-16 ***
store12	1.14108	0.05853	19.495	< 2e-16 ***
price.per.liter	-11.76280	0.44414	-26.485	< 2e-16 ***
display	-0.11574	0.03978	-2.910	0.00362 **
feature	-1.30096	0.09493	-13.704	< 2e-16 ***
coupon	0.68500	0.07347	9.323	< 2e-16 ***
PACKAGEBOTTLE:L5COKE CLASSIC	0.56665	0.25707	2.204	0.02752 *
PACKAGECAN:L5COKE CLASSIC	3.83251	0.35629	10.757	< 2e-16 ***
PACKAGEBOTTLE:L5DIET COKE	0.20505	0.25783	0.795	0.42645
PACKAGECAN:L5DIET COKE	3.65179	0.35600	10.258	< 2e-16 ***
PACKAGEBOTTLE:L5DIET PEPSI	-0.24347	0.25604	-0.951	0.34166
PACKAGECAN:L5DIET PEPSI	3.21407	0.36214	8.875	< 2e-16 ***
PACKAGEBOTTLE:L5PEPSI	0.36735	0.26101	1.407	0.15932
PACKAGECAN:L5PEPSI	3.60031	0.36442	9.880	< 2e-16 ***

Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments	3	15659	239.7	<2e-16 ***
Wu-Hausman	1	15660	506.3	<2e-16 ***
Sargan	2	NA	157.3	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.005 on 15661 degrees of freedom
Multiple R-Squared: 0.9672, Adjusted R-squared: 0.9671
Wald test: 1.942e+04 on 24 and 15661 DF, p-value: < 2.2e-16

Appendix D: IV Nested Logit (price AND ln_share_g instrumented) - MAIN MODEL

t test of coefficients:

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
ln_share_g	0.123872	0.036973	3.3503	0.0008090 ***
store01	2.023727	0.120468	16.7989	< 2.2e-16 ***
store02	0.242203	0.070712	3.4252	0.0006159 ***
store03	0.813493	0.069250	11.7472	< 2.2e-16 ***
store04	2.065598	0.131422	15.7173	< 2.2e-16 ***
store05	2.386065	0.150102	15.8963	< 2.2e-16 ***
store06	0.241941	0.060496	3.9993	6.382e-05 ***
store07	2.096145	0.175570	11.9391	< 2.2e-16 ***
store08	0.845400	0.095481	8.8542	< 2.2e-16 ***
store09	1.931360	0.176823	10.9226	< 2.2e-16 ***
store10	2.047684	0.124012	16.5120	< 2.2e-16 ***
store11	2.215623	0.129264	17.1403	< 2.2e-16 ***
store12	0.949272	0.072473	13.0983	< 2.2e-16 ***
price.per.liter	-9.641667	0.670388	-14.3822	< 2.2e-16 ***
display	-0.051351	0.035078	-1.4639	0.1432439
feature	-0.968401	0.115126	-8.4117	< 2.2e-16 ***
coupon	0.700357	0.091544	7.6505	2.118e-14 ***
PACKAGEBOTTLE:L5COKE CLASSIC	-0.289322	0.298671	-0.9687	0.3327109
PACKAGECAN:L5COKE CLASSIC	2.421269	0.466138	5.1943	2.080e-07 ***
PACKAGEBOTTLE:L5DIET COKE	-0.644381	0.297477	-2.1662	0.0303146 *
PACKAGECAN:L5DIET COKE	2.226276	0.469498	4.7418	2.137e-06 ***
PACKAGEBOTTLE:L5DIET PEPSI	-1.033567	0.283143	-3.6503	0.0002628 ***
PACKAGECAN:L5DIET PEPSI	1.834496	0.459288	3.9942	6.521e-05 ***
PACKAGEBOTTLE:L5PEPSI	-0.468713	0.295640	-1.5854	0.1128913
PACKAGECAN:L5PEPSI	2.206212	0.464537	4.7493	2.060e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>

> # Diagnostics: weak instruments, endogeneity test (Hausman), overidentification (Sargan)

> print(summary(iv_nl, diagnostics = TRUE))

Call:

```
ivreg(formula = y ~ ln_share_g - 1 + PACKAGE:L5 + store01 + store02 +
  store03 + store04 + store05 + store06 + store07 + store08 +
  store09 + store10 + store11 + store12 + price.per.liter +
  display + feature + coupon | -1 + PACKAGE:L5 + store01 +
  store02 + store03 + store04 + store05 + store06 + store07 +
  store08 + store09 + store10 + store11 + store12 + display +
  feature + coupon + avg_price_other_stores_brand + aluminium.price +
  sugar.price + aluminum_x_can + sugar_x_regular + avg_price_others_nest,
  data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.751809	-0.533101	-0.002739	0.512149	3.861657

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
ln_share_g	0.12387	0.03559	3.481	0.000502 ***
store01	2.02373	0.11851	17.077	< 2e-16 ***
store02	0.24220	0.07011	3.454	0.000553 ***
store03	0.81349	0.06698	12.145	< 2e-16 ***
store04	2.06560	0.13026	15.857	< 2e-16 ***
store05	2.38606	0.14699	16.233	< 2e-16 ***
store06	0.24194	0.05885	4.111	3.95e-05 ***
store07	2.09614	0.17132	12.235	< 2e-16 ***
store08	0.84540	0.09244	9.145	< 2e-16 ***
store09	1.93136	0.17343	11.136	< 2e-16 ***
store10	2.04768	0.12135	16.874	< 2e-16 ***
store11	2.21562	0.12847	17.246	< 2e-16 ***
store12	0.94927	0.06997	13.567	< 2e-16 ***
price.per.liter	-9.64167	0.65999	-14.609	< 2e-16 ***
display	-0.05135	0.03446	-1.490	0.136176
feature	-0.96840	0.11305	-8.566	< 2e-16 ***
coupon	0.70036	0.06255	11.197	< 2e-16 ***
PACKAGEBOTTLE:L5COKE CLASSIC	-0.28932	0.29484	-0.981	0.326467
PACKAGECAN:L5COKE CLASSIC	2.42127	0.45940	5.270	1.38e-07 ***
PACKAGEBOTTLE:L5DIET COKE	-0.64438	0.29373	-2.194	0.028264 *
PACKAGECAN:L5DIET COKE	2.22628	0.46266	4.812	1.51e-06 ***
PACKAGEBOTTLE:L5DIET PEPSI	-1.03357	0.28022	-3.688	0.000226 ***
PACKAGECAN:L5DIET PEPSI	1.83450	0.45411	4.040	5.38e-05 ***
PACKAGEBOTTLE:L5PEPSI	-0.46871	0.29217	-1.604	0.108676
PACKAGECAN:L5PEPSI	2.20621	0.45832	4.814	1.50e-06 ***

Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments (ln_share_g)	6	15656	769.0	<2e-16 ***
Weak instruments (price.per.liter)	6	15656	168.3	<2e-16 ***
Wu-Hausman	2	15658	353.1	<2e-16 ***
Sargan	4	NA	199.4	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8732 on 15660 degrees of freedom

Multiple R-Squared: 0.9752, Adjusted R-squared: 0.9752

Wald test: 2.473e+04 on 25 and 15660 DF, p-value: < 2.2e-16

Appendix E: OLS Nested Logit Elasticities

Product shares (OLS):

PACKAGEBOTTLE:L5COKE CLASSIC

s.cc.b	0.004806021
s.dc.b	0.003279144
s.p.b	0.003469717
s.dp.b	0.002085497
s.cc.c	0.008801547
s.dc.c	0.007349464
s.p.c	0.005140388
s.dp.c	0.003772215
s.0	0.961296005

Own-price elasticities (OLS):

price.per.liter

el.cc.b	-2.745443
el.dc.b	-2.791405
el.p.b	-2.959730
el.dp.b	-2.999202
el.cc.c	-3.197056
el.dc.c	-3.049858
el.p.c	-3.866232
el.dp.c	-3.843038

Cross-price elasticities within group (OLS):

price.per.liter

elc.cc.b	0.4844704
elc.dc.b	0.4452399
elc.p.b	0.3594288
elc.dp.b	0.2875513
elc.cc.c	1.2108182
elc.dc.c	1.3594565
elc.p.c	0.7387846
elc.dp.c	0.7224785

Cross-price elasticities across groups (OLS):

price.per.liter

elcc.cc.b	0.004867367
elcc.dc.b	0.003327922
elcc.p.b	0.003611102
elcc.dp.b	0.002149287
elcc.cc.c	0.012164823
elcc.dc.c	0.010161185
elcc.p.c	0.007422405
elcc.dp.c	0.005400127

OLS Nested Logit Elasticities:

	price.per.liter	price.per.liter	price.per.liter
el.cc.b	-2.745443	0.4844704	0.004867367
el.dc.b	-2.791405	0.4452399	0.003327922
el.p.b	-2.959730	0.3594288	0.003611102
el.dp.b	-2.999202	0.2875513	0.002149287
el.cc.c	-3.197056	1.2108182	0.012164823
el.dc.c	-3.049858	1.3594565	0.010161185
el.p.c	-3.866232	0.7387846	0.007422405
el.dp.c	-3.843038	0.7224785	0.005400127

Within group share:

PACKAGEBOTTLE:L5COKE CLASSIC	
s.cc.b.g	0.2175572
s.dc.b.g	0.2002886
s.p.b.g	0.1516318
s.dp.b.g	0.1213300
s.cc.c.g	0.4064129
s.dc.c.g	0.4561684
s.p.c.g	0.2243981
s.dp.c.g	0.2222130

Appendix F: IV Nested Logit Elasticities

```
print(paste("Sigma (IV):", round(sigma, 4)))  
[1] "Sigma (IV): 0.1239"  
> print(paste("Alpha (IV):", round(alpha, 4)))  
[1] "Alpha (IV): -9.6417"
```

Within-group shares (IV):" (IV):

PACKAGEBOTTLE:L5COKE CLASSIC	
s.cc.b.g	0.2201533
s.dc.b.g	0.2030089
s.p.b.g	0.1489035
s.dp.b.g	0.1172668
s.cc.c.g	0.4155212
s.dc.c.g	0.4650752
s.p.c.g	0.2154220
s.dp.c.g	0.2146491

```
"Group shares (IV):"  
> print(paste("Group 1 (REGULAR):", round(s.g1, 4)))
```

```
[1] "Group 1 (REGULAR): 0.0076"
> print(paste("Group 2 (DIET):", round(s.g2, 4)))
[1] "Group 2 (DIET): 0.0056"
```

Product shares:

s.cc.b	0.0016674879
s.dc.b	0.0011432864
s.p.b	0.0011278263
s.dp.b	0.0006604123
s.cc.c	0.0031472452
s.dc.c	0.0026191675
s.p.c	0.0016316518
s.dp.c	0.0012088407
s.0	0.9867940818

Own-price elasticities (IV):

el.cc.b	-6.547732
el.dc.b	-6.578826
el.p.b	-6.793070
el.dp.b	-6.756442
el.cc.c	-8.701137
el.dc.c	-8.651747
el.p.c	-9.341297
el.dp.c	-9.265614

Cross-price elasticities within group (IV):

PACKAGEBOTTLE:L5COKE CLASSIC

elc.cc.b	0.1936936
elc.dc.b	0.1766479
elc.p.b	0.1346269
elc.dp.b	0.1036193
elc.cc.c	0.4989090
elc.dc.c	0.5513060
elc.p.c	0.2702217
elc.dp.c	0.2634616

Cross-price elasticities across groups (IV):

PACKAGEBOTTLE:L5COKE CLASSIC

elcc.cc.b	0.009848766
elcc.dc.b	0.006766721
elcc.p.b	0.006845394
elcc.dp.b	0.003969269
elcc.cc.c	0.025368103
elcc.dc.c	0.021118477
elcc.p.c	0.013740005
elcc.dp.c	0.010092233

>

IV Nested Logit Elasticities:

	PACKAGEBOTTLE:L5COKE	CLASSIC	PACKAGEBOTTLE:L5COKE	CLASSIC
PACKAGEBOTTLE:L5COKE	CLASSIC			
el.cc.b	-6.547732	0.1936936	0.009848766	
el.dc.b	-6.578826	0.1766479	0.006766721	
el.p.b	-6.793070	0.1346269	0.006845394	
el.dp.b	-6.756442	0.1036193	0.003969269	
el.cc.c	-8.701137	0.4989090	0.025368103	
el.dc.c	-8.651747	0.5513060	0.021118477	
el.p.c	-9.341297	0.2702217	0.013740005	
el.dp.c	-9.265614	0.2634616	0.010092233	

```
> print(paste("Alpha (IV):", round(alpha_iv, 4)))
[1] "Alpha (IV): -9.6417"
> print(paste("Sigma (IV):", round(sigma_iv, 4)))
[1] "Sigma (IV): 0.1239"
> print(paste("Mean own-price elasticity (IV):", round(mean_elasticity, 4)))
[1] "Mean own-price elasticity (IV): -7.8295"
> print(paste("Ratio of within/across nest cross-elasticities (sigma/(1-sigma)):", round(ratio, 4)))
[1] "Ratio of within/across nest cross-elasticities (sigma/(1-sigma)): 0.1414"
```